Week-8 (DAY-5) – Handson – Keerthivasan R

📌 Problem Statement

In modern software development, applications must be reliable, maintainable, and error-free. However, many developers deploy applications without proper testing, which leads to:

- Unexpected runtime errors
- Business logic failures
- Incorrect calculations
- Security vulnerabilities
- Difficult debugging and maintenance

To ensure software quality, automated unit testing must be implemented.

This project aims to implement **unit testing using NUnit framework** in a .NET / ASP.NET Core application to validate business logic and ensure the correctness of application components.

🎯 Objective

Develop and test a sample ASP.NET Core application using **NUnit** to:

- Verify business logic functionality
- Test service layer methods
- Validate input and output conditions
- Handle exception testing
- Improve code reliability

📐 System Description

Consider a simple **Banking / Insurance / Employee Management System** with the following operations:

- Add Employee / Customer
- Calculate Salary / Premium
- Process Claims
- Perform Deposit / Withdrawal
- Generate Reports

The goal is to write **NUnit test cases** for:

1. Valid input scenarios
2. Invalid input scenarios
3. Boundary value testing
4. Exception handling
5. Business rule validation

⚙️ Technical Requirements

- .NET 6 / .NET 7 / .NET 8
- NUnit Framework
- NUnit3TestAdapter
- Microsoft.NET.Test.Sdk
- Visual Studio

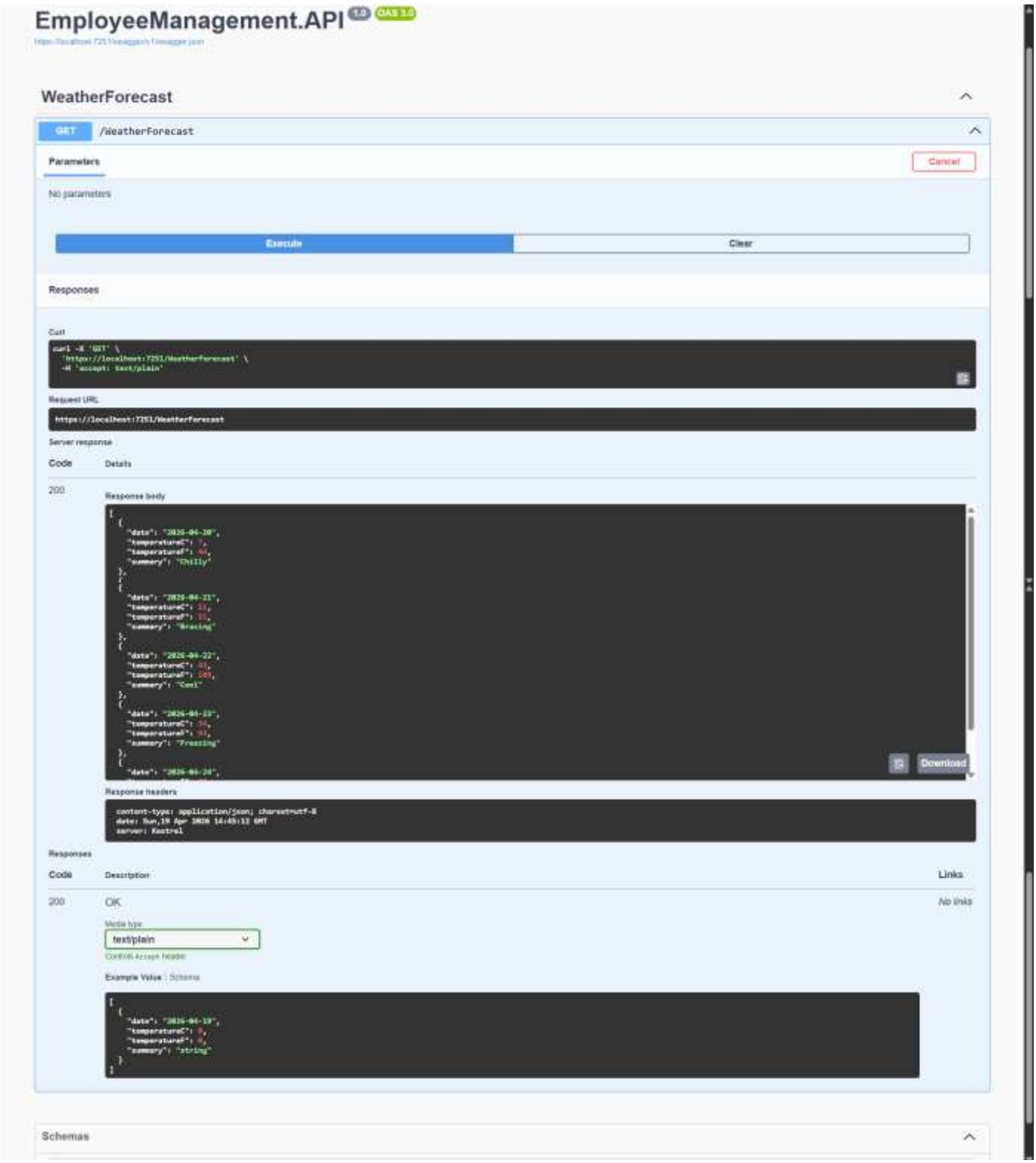
🧩 Example Testing Scenarios

- Test if salary calculation returns correct value
- Test if withdrawal fails when balance is insufficient
- Test if claim amount cannot be negative
- Test if exception is thrown for null inputs
- Test API response status codes

🚀 Expected Outcome

- All business logic is validated through automated test cases
- Bugs are identified early in development
- Code becomes maintainable and refactor-friendly
- Application reliability increases

OUTPUT:


EmployeeManagement.API
WeatherForecast
GET
/WeatherForecast
Parameters
Cancel
No parameters
Execute
Clear
Responses
Curl
curl -X 'GET' \
'https://localhost:7251/WeatherForecast' \
-H 'accept: text/plain'
Request URL
https://localhost:7251/WeatherForecast
Server response
Code
Details
200
Response body
Download
Response headers
content-type: application/json; charset=utf-8
date: Sun,19 Apr 2026 14:45:12 GMT
server: Kestrel
Responses
Code
Description
Links
200
OK
No links
Media type
text/plain
Example Value | Schema
Schemas